p-adic Representation Theory

— *EYNTKA* Series —

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Contents

0.0.1 Inspiration Very heavily inspired by [2]

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Smooth Representations

here

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [4, chapter 8]. Here is a list of useful facts to remember:

- The topology of G is translation invariant: If $U \subseteq G$ is open, then Ux and xU is open
- If $H \leqslant G$, G/H has the quotient topology, and if $H \trianglelefteq G$ then G/H is a topological group.
- if $H \leqslant G$ is open it must be closed hence connected ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the compliment of H). More strictly, if H is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open U, then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If G is connected, it has no non-trivial proper open subgroups. We shall heavily be working with totally disconnected groups.
- Show that $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Note that $\mathbb{Q}_p$ is totally disconnected, hence this does not contradict the above remark. For a review of p-adic integers, see [5, chapter 1.1].
- A closed subgroup $H \leqslant G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). If G is first-countable, H is closed if it contains all its limit points (more generally, it it contains all the limits of nets).
- A subgroup need not be open or closed: $\mathbb{Q} \subseteq \mathbb{R}$.
- For every open neighborhood of e, say for example U, there is a symmetric neighborhood (if $A \subseteq G$ such that $A = A^{-1}$, we'll say A is *symmetric*) V of e such that $V \subseteq U$
- For every neighborhood U of e, there is a neighborhood V of e with $VV \subseteq U$
- if H is a subgroup of G, so it $\overline{H}$ (the closure under limits/nets).

- Every open subgroup of G is also closed
- If K_1, K_2 are compact subset of G, so is $K_1 K_2$

Lemma 1.0.1: Characterizing Hausdorffness

Let G be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$ Then:

1. H is a subgroup
2. H is the closure of e, i.e. $\overline{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

***Proof*:**
exercise

Henceforth, we shall always assume Hausdorffness unless specified otherwise.

1.1 Locally Profinite Groups

Definition 1.1.1: Profinite Group

Take a diagram finite groups with the discrete topology. Then a *profinite group* is a topological group that is the colimit of the system. Given any topological group G, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \to \widehat{G}$ is a continuous morphism and G is dense in $\widehat{G}$.

Example 1.1.2: Profinite Groups
We have the following "trivial" examples:

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the "usual" example and is better suited for the intuition of profinite groups. Let

$G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \cdots$) the element mod $p^k\mathbb{Z}$ eventually stabilize. For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005$ mod 7^n for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

Another important non-abelian example is the profinite completion of galois Galois groups, namely $\mathrm{Gal}_K(\overline{K})$. See [6] or a quick overview or [1] for more details.

More generally, profinite groups can be seen as the object that captures all the finite behaviour "up to infinity" (or philosophically, all the "potential infinity") of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each p^n and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group, see ref:HERE.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both.

Lemma 1.1.3: Pro-finite Completion Injective

Let G be a topological group and $\widehat{G}$ it's profinite completion. Then $G \to \widehat{G}$ is injective if the intersection all finite-index open normal subgroups of G is trivial

***Proof*:**
exercise.

The completion comes equipped with a natural universal property:

Theorem 1.1.4: Universal Property of Profinite Completion

let $\varphi : G \to H$ be continuous homomorphism with H a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

***Proof*:**
exercise (hint: recall G is dense in $\widehat{G}$).

A striking result is that all profinite groups can be characterized by a simple topological property:

Theorem 1.1.5: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
Let G be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.

Conversely, suppose G is totally disconnected and compact. We must show it comes from an inverse system of finite groups. will finish later.

Corollary 1.1.6: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all *open* normal subgroups ordered by contain need.
If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

***Proof*:**
exercise.

Definition 1.1.7: Locally Profinite Group

A topological gorp G is *locally profinite* it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of G.

Notice how there is no mention of an inverse system. The reader should show that the definition imples G is totally disconnected, and hence if it is also compact then we recover a profinite group. Every discrete group is sub finite, closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite subgroup is locally compact. It is an exercise to show that any locally compact totally disconnected group is locally profinite. The canonical example for these notes of a locally profinite group is the non-Archimedean local field $(F, +)$ as proven in [5]. Similarly, $(F^\times, \times)$ is a profinite group by passing through e^x, or:

$$U_F^n = 1 + \mathfrak{p}^n$$

Being profinite is closed under products, hence F^n is profinite, and $(M_n(F), +)$ is locally profinite under addition. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ under multiplication is an open topological subgroup. It is locally profinite as $\mathrm{GL}_n(\mathcal{O}_F)$ is profinite (it is compact as it is homeomorphic to a closed subset of $\mathcal{O}_F^{n^2}$ which is a product of compact groups, and it is open as $\mathcal{O}_F = \{x \in F \mid |x|_\mathfrak{p} \leq 1\}$ is an open set. Note that this was all done in coordinates, more generally this applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$.

Let us now consider character's on locally profinite groups:

Lemma 1.1.8: Continuous Character

let $\varphi : G \to \mathbb{C}^\times$ be a group homomorphism. Then the following are equivalent:

1. φ is continuous
2. the kernel of φ is open

If φ satisfies one of these conditions and G is the union of its compact open subgroups ,then $\mathrm{im}\psi \subseteq S^1 \subseteq \mathbb{C}^\times$.

***Proof*:**
If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker\varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing it is open. Conversely, Let φ be continuous and U an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any U, we can pick U sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^\times$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker\varphi$. As K is open, $\ker\varphi$ contains an open subgroup and hence by topological group theory $\ker\varphi$ is open.

Next, S^1 is the unique maximal compact subgroup of $\mathbb{C}^\times$. If K' is a compact subgroup of G, $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

Definition 1.1.9: Continuous Character

A *character* for a locally profinite group G is a continuous homomorphism

$$\varphi : G \to \mathbb{C}^\times$$

We shall often write 1_G or just 1 for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, φ is called a *unitary character*.

summary of next section On page 12 in [2], he defines $\widehat{F}$ to be the set of all characters of F. As F is a local field, it is the union of its compact open subgroup $\mathfrak{p}^n$ for $n \in \mathbb{N}$, hence all characters are unitary. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of φ .Fixing a d, the set of characters of level $\leq d$ is a subgropu.This defines a duality which extends to the multiplicative set. A similar thing has already been done in my CFT notes [5] in less generality, so skipping for now

1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

Example 1.2.1: "Useless" Representation

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$. By field theory, it is isomorphic to $\mathbb{C}$. Thus there is an injective ring homomorphism $\mathbb{Q}_p \to \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all the important information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the representation of topological groups. Often when G is not compact, we get a vector space V that is infinite dimensional with which is either a Banach or Hilbert space and $\mathrm{GL}(V)$ is the set of all invertible bounded linear operators; this thread is further explored in [3].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness is different from smoothness for manifolds as we will be looking at maps $\mathbb{Q}_p \to \mathbb{C}$ which don't share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in close will get the same linear, or "constant" if we change coordinates in our perspective. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

Definition 1.2.2: Smooth Map For Locally Profinite Groups

Let G be a locally profinite group. Then a map $f : G \to \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood U containing x such that $f(x) = f(y)$ for all $x, y \in U$.

1.2.3 Precision in Vocabulary Via Harmonic Analysis There is a more precise comparison when considering the Fourier transform with distributions, so we'll assume [4] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of "ideal" functions is the Schwartz space, consisting of functions that are both smooth (C^∞) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.

2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis. The "ideal" space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support. The Fourier transform on Q_p is a perfect map on this space: the transform of a locally constant, compactly supported function is another locally constant, compactly supported function. See this link for more details.

Proposition 1.2.4: Equivalence Notion of Smoothness

Let G be a locally profinite group and $f : G \to \mathbb{C}^n$ a map. Then f is smooth if and only if for *any subset* $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if f is smooth, it is continuous.

***Proof*:**
exercise.

The key in the notion of smoothness being useful is that on $\mathbb{Q}_p$ there are non-trivial locally constant functions:

Example 1.2.5: Smooth Functions On Locally Profinite Groups

1. The p-adic valuation $|-|_p : \mathbb{Q}_p^\times \to \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of p^{-n} is $p^n\mathbb{Z}_p^\times$ which is open.

   Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretcehd to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

By identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$, we get the notion of a smooth representation:

Definition 1.2.6: Smooth Representation

Let G be a locally profinite group. Then a *representation* of G is a pair (π, V) where V is a complex vector space and $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation (π, V) is called *smooth* if for every $v \in V$, there is a compact open subgroup $K_v \subseteq G$ such that for all $k \in K_v$:

$$\pi(k) \cdot v = v$$

Equivalently, if V^K denotes the space of $\pi(K)$-fixed vector in V,

$$V = \bigcup_K V^K$$

where K ranges over all compact open subgroups of G.

Proposition 1.2.7: Smooth Representation Alternative

A represetnation $\pi : G \to \mathrm{GL}(V)$ is smooth if

$$\mathrm{Stab}_G(v) = \{g \in G \mid \pi(g) \cdot v = v\}$$

is an open subgroup of G for all $v \in V$

***Proof*:**
exercise.

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